WEEK 6 - WEB Scraping - web scaping exercise using python

business data analytic (Universiti Teknologi Malaysia)

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WEEK 5 – WEBSCRAPING

Web Scraping using Python

Covers:

1. Google Colab Notebook.
2. Install library Beautiful Soup.
3. Identify element in webpage.
4. Scrape an element.

Instructions :

1. Google Colab
   - Using the link : https://colab.research.google.com/
   - Click on 'New Notebook'
2. Importing Libraries
   - The first line import the *urlopen* function from the *urllib.request* module and rename it to *uReq* for convenience.
   - Then, the BeautifulSoup class is imported from the *bs4* (Beautiful Soup 4) module and renamed as *soup*.

```
from urllib.request import urlopen as uReq
from bs4 import BeautifulSoup as soup
```

3. Defining the URL:
   - This line defines the URL of the webpage that the user wishes to scrape from.

```
my_url = 'https://webscraper.io/test-sites/e-commerce/allinone'
```

4. Opening the Connection and Reading the Page:
   - These lines open a connection to the URL using urlopen(), reads the HTML content of the page, and then closes the connection.

```
uClient = uReq(my_url)
page_html = uClient.read()
uClient.close()
```

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5. Parsing the HTML Content:

   - This line parses the HTML content of the page using BeautifulSoup's soup() function. We specify the HTML parser ("html.parser") as the second argument.

```
page_soup = soup(page_html, "html.parser")
```

6. Looping Through Items

   - This line loops through all the elements with the class "card" on the page. Each item represents one product card on the webpage.

```
i = 0
for item in page_soup.findAll("div", {"class": "card"}):
    i += 1
    print(i)
```

7. Extracting the Information

   - This line (item.find) finds the specific element (h4, a, and p) within the current item that has the specified class attributes;

```
    price_element = item.find("h4", {"class": "price float-end card-title
pull-right"})
    if price_element:
        price = price_element.text.strip()
        print("Price:", price)

    title = item.find("a", {"class": "title"})
    if title:
        thetitle = title['title']
        print("Title:", thetitle)

    desc = item.find("p", {"class": "description card-text"}).text.strip()
    print("Description:", desc)

    rating_element = item.find("p", {"class": "review-count float-end"})
    if rating_element:
        rating = rating_element.text.strip()
        print("Rating:", rating)

    print("\n")
```

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Task:

1. Open https://colab.research.google.com/
2. Create new notebook
3. Import BeautifulSoup library

```
from bs4 import BeautifulSoup
import requests
```

4. Open webpage: https://en.wikipedia.org/wiki/List_of_largest_companies_in_the_United_States_by_revenue

5. Request link to the url

```
# Open a connection to the URL and read its HTML content
uClient = uReq(wiki_url)
page_html = uClient.read()
uClient.close()
```

6. Study the elements of the webpage.
   - Scrape the table of 'List of the largest public / publicly traded companies' column name only.

```
from urllib.request import urlopen as uReq
from bs4 import BeautifulSoup as soup

wiki_url = 'https://en.wikipedia.org/wiki/List_of_largest_companies_in_the_United_States_by_revenue'

uClient = uReq(wiki_url)
page_html = uClient.read()
uClient.close()

page_soup = soup(page_html,"html.parser")

table = page_soup.find("table",{"class": "wikitable"})

column_names = [header.text.strip() for header in table.find_all("th")]

print(column_names)
```

```
['Rank', 'Name', 'Industry', 'Revenue (USD millions)', 'Revenue growth', 'Employees', 'Headquarters']
```

   - Scrape the whole table.

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```
[4] from urllib.request import urlopen as uReq
    from bs4 import BeautifulSoup as soup

    wiki_url = 'https://en.wikipedia.org/wiki/List_of_largest_companies_in_the_United_States_by_revenue'

    uClient = uReq(wiki_url)
    page_html = uClient.read()
    uClient.close()

    page_soup = soup(page_html,"html.parser")
    table = page_soup.find("table",{"class": "wikitable"})

    if table:
      rows = table.find_all("tr")
      headers = [header.text.strip() for header in rows[0].find_all("th")]

      print(headers)

      for row in rows[1:]:
        cols = [col.text.strip() for col in row.find_all("td")]
        if cols:
          print(cols)
```

```
['Rank', 'Name', 'Industry', 'Revenue (USD millions)', 'Revenue growth', 'Employees', 'Headquarters']
['1', 'Walmart', 'Retail', '648,125', '6.0%', '2,100,000', 'Bentonville, Arkansas']
['2', 'Amazon', 'Retail and cloud computing', '574,785', '11.9%', '1,525,000', 'Seattle, Washington']
['3', 'Apple', 'Electronics industry', '383,482', '-2.8%', '161,000', 'Cupertino, California']
['4', 'UnitedHealth Group', 'Healthcare', '371,622', '14.6%', '440,000', 'Minnetonka, Minnesota']
['5', 'Berkshire Hathaway', 'Conglomerate', '364,482', '20.7%', '396,500', 'Omaha, Nebraska']
['6', 'CVS Health', 'Healthcare', '357,776', '10.9%', '259,500', 'Woonsocket, Rhode Island']
['7', 'ExxonMobil', 'Petroleum industry', '344,582', '-16.7%', '61,500', 'Spring, Texas']
['8', 'Alphabet', 'Technology and cloud computing', '307,394', '8.7%', '182,502', 'Mountain View, California']
['9', 'McKesson Corporation', 'Health', '276,711', '4.8%', '48,000', 'Irving, Texas']
['10', 'Cencora', 'Pharmacy wholesale', '262,173', '9.9%', '44,000', 'Conshohocken, Pennsylvania']
['11', 'Costco', 'Retail', '242,290', '6.8%', '316,000', 'Issaquah, Washington']
['12', 'JPMorgan Chase', 'Financial services', '239,425', '54.7%', '309,926', 'New York City, New York']
['13', 'Microsoft', 'Technology and cloud computing', '211,915', '6.9%', '221,000', 'Redmond, Washington']
['14', 'Cardinal Health', 'Healthcare', '205,012', '13.0%', '47,520', 'Dublin, Ohio']
['15', 'Chevron Corporation', 'Petroleum industry', '200,949', '-18.4%', '45,600', 'Houston, Texas']
['16', 'Cigna', 'Health insurance', '195,265', '8.2%', '71,413', 'Bloomfield, Connecticut']
['17', 'Ford Motor Company', 'Automotive industry', '176,191', '11.5%', '177,000', 'Dearborn, Michigan']
['18', 'Bank of America', 'Financials', '171,912', '49.4%', '212,985', 'Charlotte, North Carolina']
['19', 'General Motors', 'Automotive industry', '171,842', '9.6%', '163,000', 'Detroit, Michigan']
['20', 'Elevance Health', 'Healthcare', '171,340', '9.4%', '104,900', 'Indianapolis, Indiana']
['21', 'Citigroup', 'Financials', '156,820', '55.1%', '237,925', 'New York City, New York']
['22', 'Centene', 'Healthcare', '153,999', '6.5%', '67,700', 'St. Louis, Missouri']
['23', 'The Home Depot', 'Retail', '152,669', '-3.0%', '463,100', 'Atlanta, Georgia']
['24', 'Marathon Petroleum', 'Petroleum industry', '150,307', '-16.5%', '18,200', 'Findlay, Ohio']
['25', 'Kroger', 'Retail', '150,039', '-3.6%', '414,000', 'Cincinnati, Ohio']
['26', 'Phillips 66', 'Petroleum industry', '149,890', '-14.7%', '14,000', 'Houston, Texas']
['27', 'Fannie Mae', 'Financials', '141,240', '16.2%', '8,100', 'Washington, D.C.']
['28', 'Walgreens Boots Alliance', 'Pharmaceutical industry', '139,081', '4.8%', '268,500', 'Deerfield, Illinois']
['29', 'Valero Energy', 'Petroleum industry', '139,001', '-18.8%', '9,987', 'San Antonio, Texas']
['30', 'Meta Platforms', 'Technology', '134,902', '15.7%', '67,317', 'Menlo Park, California']
['31', 'Verizon Communications', 'Telecommunications', '133,974', '-2.1%', '105,400', 'New York City, New York']
```

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